Supplemental Material

Physiologically based pharmacokinetic models to predict maternal pharmacokinetics and fetal exposure to emtricitabine and acyclovir

Development of PBPK models

Acyclovir

Physicochemical properties of acyclovir were loaded from a template provided with the software.[1, 2] This template was involved only unchanged renal excretion, it was extended with acyclovir biotransformation to CMMG. Since ALDH 2 is predominantly expressed in the liver,[3] this biotransformation was implemented as hepatic plasma clearance process. In the non-pregnant PBPK model, specific tubular secretion and specific hepatic plasma clearance were simultaneously fitted to *in vivo* disposition profiles in non-pregnant subjects following IV administration of different doses[4] and to the dose fraction excreted unchanged in urine given in Table 1. Thereafter, the model was used to simulate the PK following oral administration in non-pregnant women. The initial non-pregnant oral PBPK model underestimated the peak plasma concentration (C_{max}) because of low acyclovir absorption. Therefore, the oral PBPK model was refined by fitting the intestinal permeability to the *in vivo* PK data.[5] Once the non-pregnant model described the *in vivo* data adequately, it was translated to pregnancy following the same procedure as described below for emtricitabine. No information could be found as to whether pregnancy is associated with alterations in ALDH 2 activity, which was thus assumed to remain unchanged in the pregnancy oral PBPK model.

Emtricitabine

Total body clearance was implemented as glomerular filtration and first-order tubular secretion with a minor contribution from hepatic clearance. In the non-pregnant PBPK model, intestinal permeability, specific tubular secretion rate, and specific hepatic plasma clearance were

simultaneously fitted to the dose fraction excreted unchanged in urine given in Table 1 and to the plasma concentration-time profiles that were reported in the comparison studies.[6-8] Once the non-pregnant model described the *in vivo* data adequately, it was translated to pregnancy following a previously described workflow.[9] The physicochemical properties listed in Table 1 were kept constant in the pregnancy PBPK model and the fraction unbound listed in Table 1 was estimated to pregnancy using a previously described scaling approach.[10] Of note, the specific clearance values (absolute clearance divided by organ volume) were also kept constant, but due to an increase in kidney volume and renal blood flow, the absolute tubular secretion rate was higher in the pregnancy than in the non-pregnant model. Elimination in the fetus was not accounted for in the model. According to the study protocol, 11/26 patients were instructed to take a light meal with their medications. The remaining of patients had no food intake restriction. Therefore, a light meal intake was applied into the emtricitabine model according to available information in the study protocol (meal energy content ~360 kcal).

Table S1: Summary of clinical pregnancy studies used for model evaluation

	Emtricitabine	Acyclovir
Source	IMPAACT P1026s[11] and Published PK study [12]	Published PK studies [13-15]
Dose (mg)	200	400
Frequency	Once daily	Single dose and multiple dose (t.i.d.)
Route of administration	Oral	Oral
N	24, 33	10
Study design	Prospective pharmacokinetic study of HIV-infected pregnant women taking antiretrovirals for clinical indications	Pharmacokinetic studies
Gestational age (weeks)	23 – 40 [a]	36 (single dose); 38 (multiple dose)
Weight (kg)	50.8 – 137 [a]	60 – 89[b]

[a] Data from IMPAACT P1026s; no range reported by Mendes et al. [12]

[b] Data from Frenkel et al.[13]; no range reported by Kimberlin et al[14] and Leung et al.[15]

Supplementary Figures:

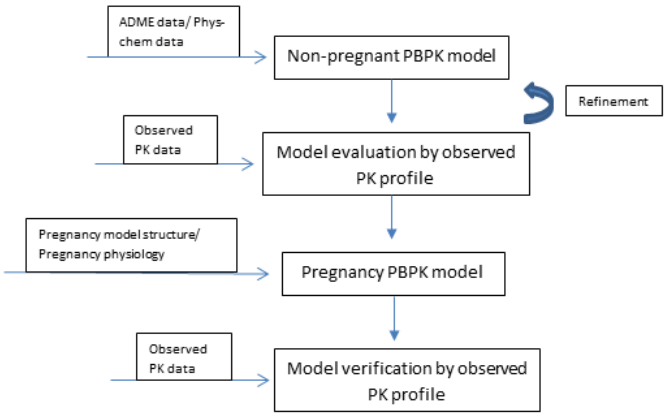


Figure S1: Workflow for building the PBPK pregnancy model of emtricitabine and acyclovir

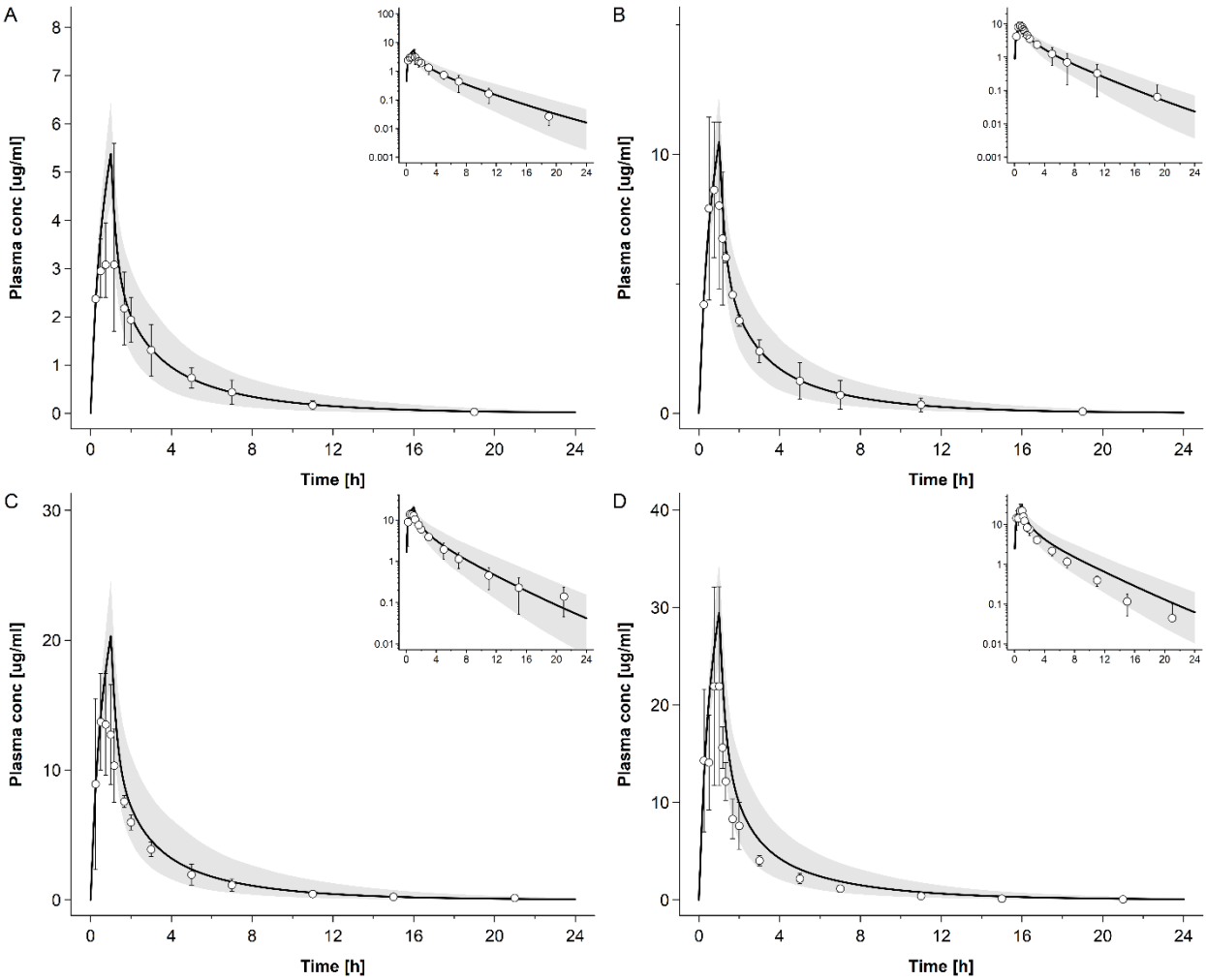


Figure S2: Plasma concentration-time profiles of acyclovir following intravenous administration of different doses in non-pregnant subjects. Circles represent observed *in vivo* data from Laskin et al;[4] the solid line represents the simulated plasma concentration and the shaded area represents the simulated 5th – 95th percentile range. (A: weight: 86.2+/-32.1 kg; B: weight: 86.2+/-32.1 kg; C: weight: 73.3+/-16.5 kg; D: weight: 63.7+/-18.5 kg). Semi-log scale figures are given as inset figure in the top right corners. **A**: IV 2.5mg/kg; **B**: IV 5mg/kg; **C**: IV 10mg/kg; **D**: IV 15mg/kg.

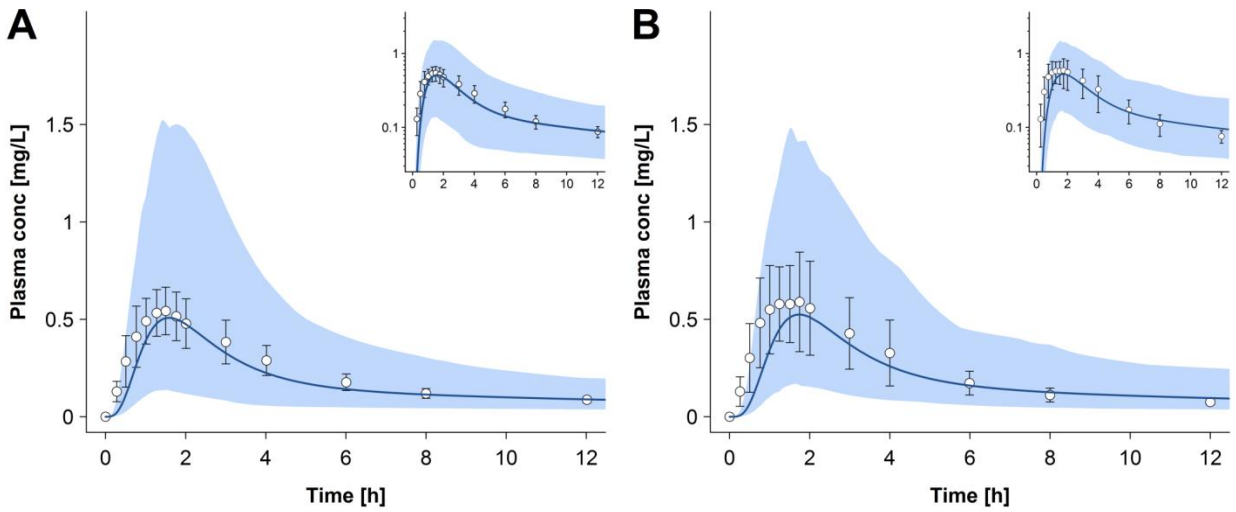


Figure S3: Plasma concentration-time profiles of acyclovir following single oral administration of 400 mg as suspension or tablet in non-pregnant subjects. Circles represent observed *in vivo* data and error bars the observed standard deviation from Palma-Aguirre et al;[5] the solid line represents the simulated mean plasma concentration and the shaded area represents the simulated 5th – 95th percentile range. Semi-log scale figures are given as inset figure in the top right corners. **A**: 400 mg as oral suspension; **B**: 400 mg as oral tablet.

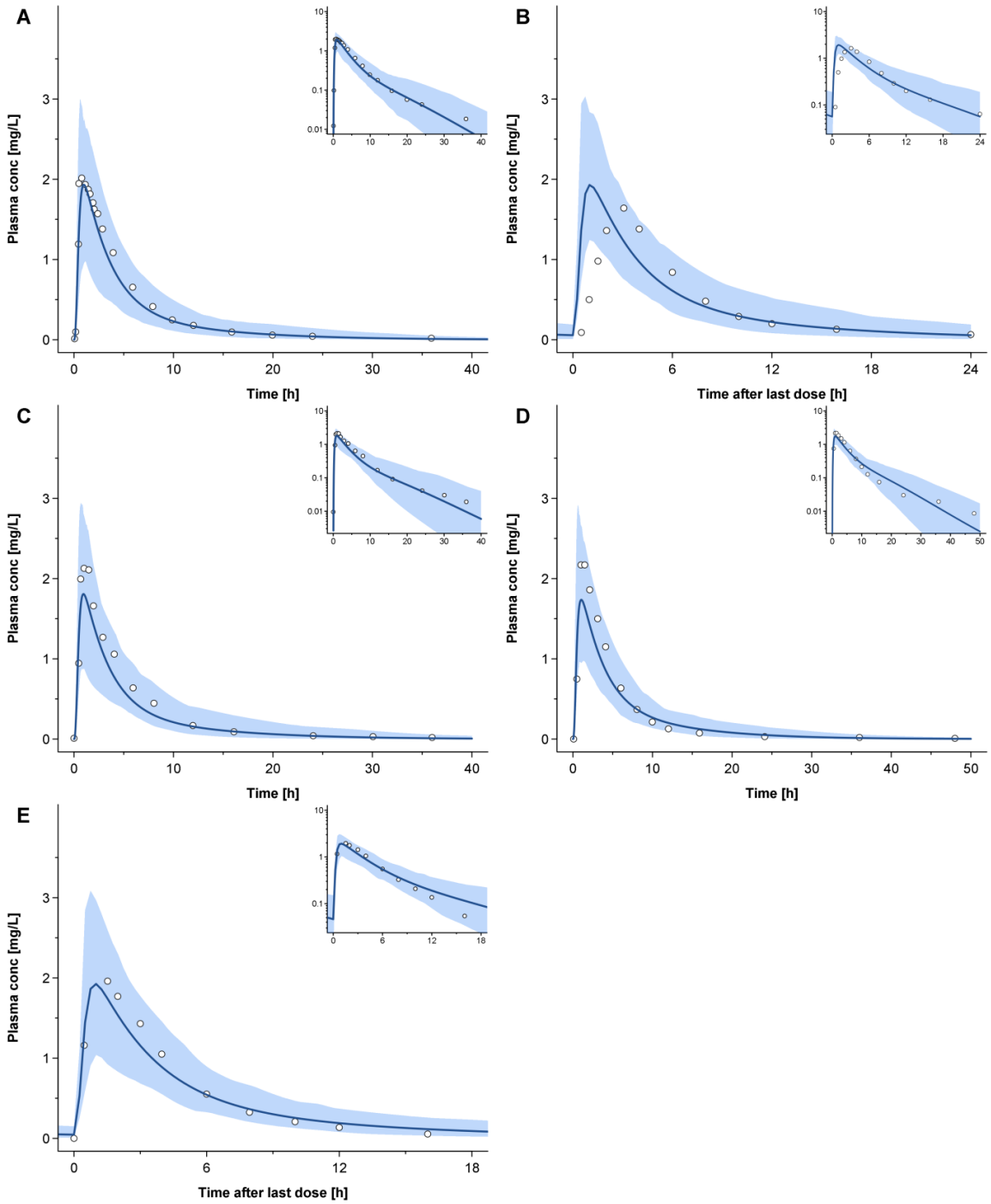


Figure S4: Plasma concentration-time profiles of emtricitabine following single and multiple oral administration of 200 mg in non-pregnant subjects. Circles represent observed *in vivo* data from prior publications;[6-8] the solid line represents the simulated mean plasma concentration and the shaded area the simulated or predicted 5th – 95th percentile range. Semi-log scale figures are given as inset figure in the top right corners. **A**: single dose oral 200 mg tablet; **B**: multiple dose oral tablet 200 mg once a day in steady-state; **C**: single dose oral tablet 200 mg; **D**: single dose oral tablet 200 mg; **E**: multiple dose oral tablet 200 mg once a day in steady-state.

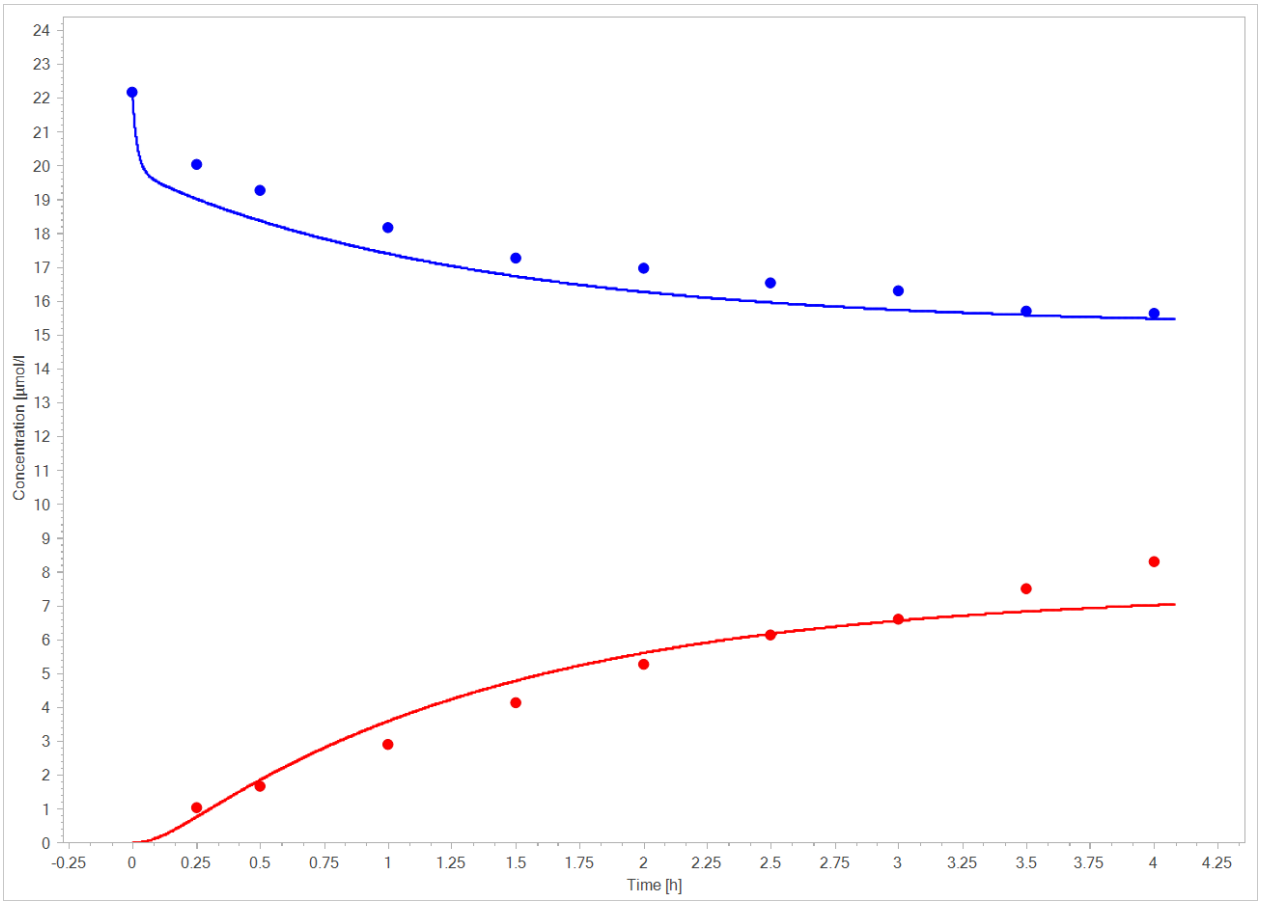


Figure S5: Concentration-time profile of acyclovir in the *ex vivo* cotyledon perfusion experiment. The circles indicate the measured acyclovir concentrations in the maternal (blue) and fetal (red) chamber taken from Henderson et al[16] and the lines represent the simulated concentrations in the maternal (blue) and fetal (red) chamber.

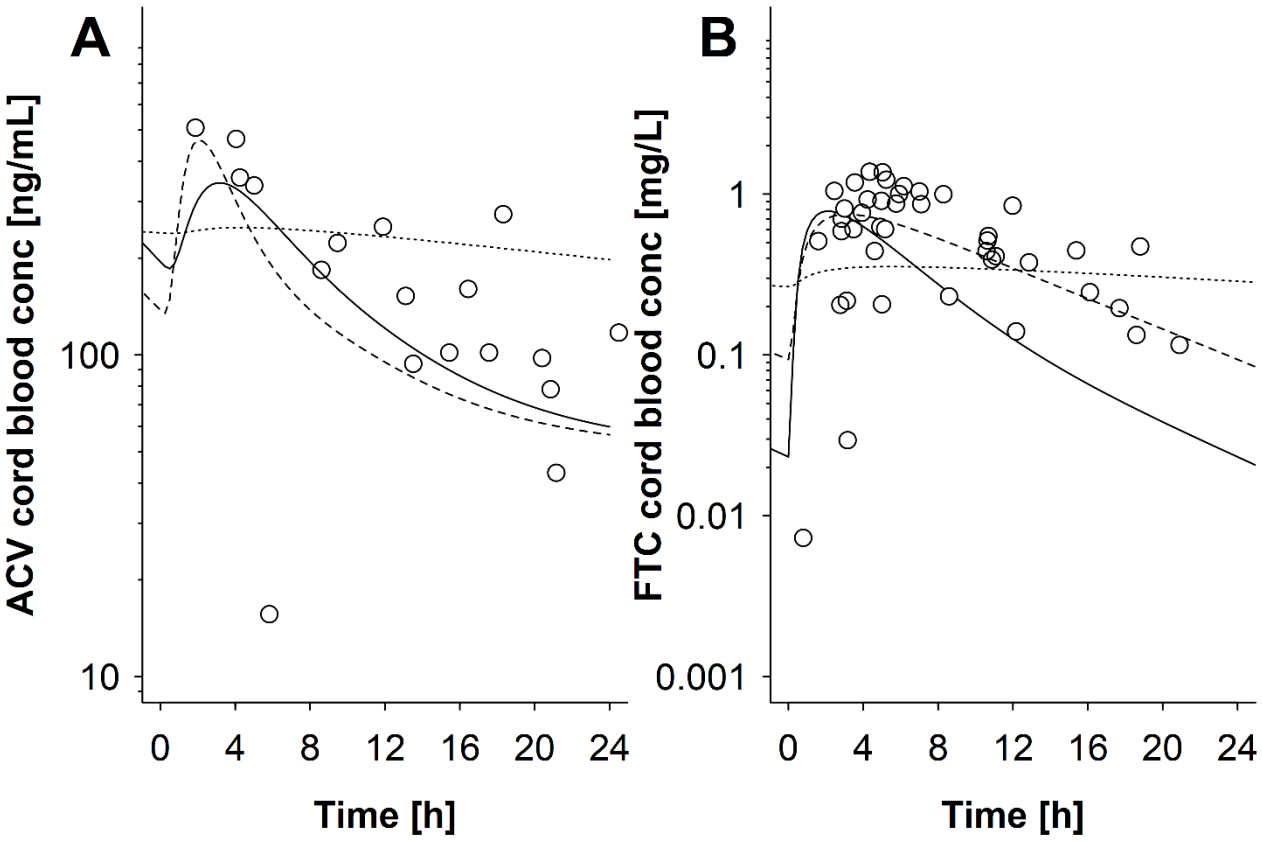
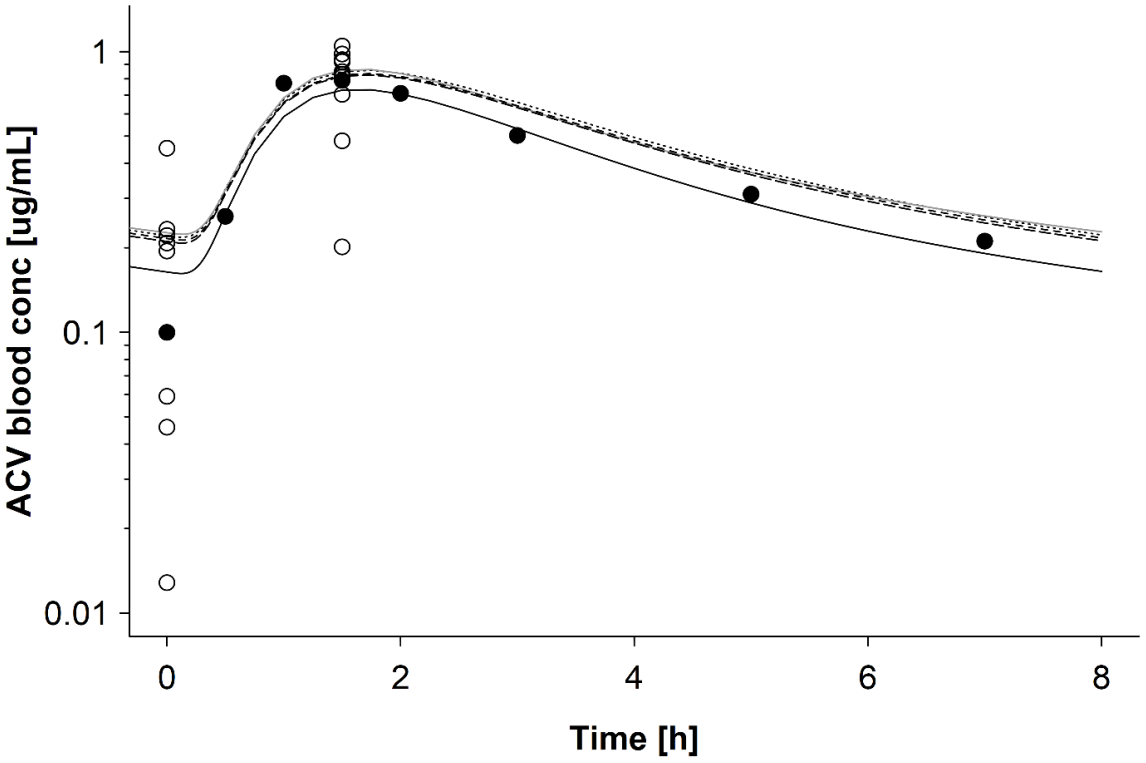
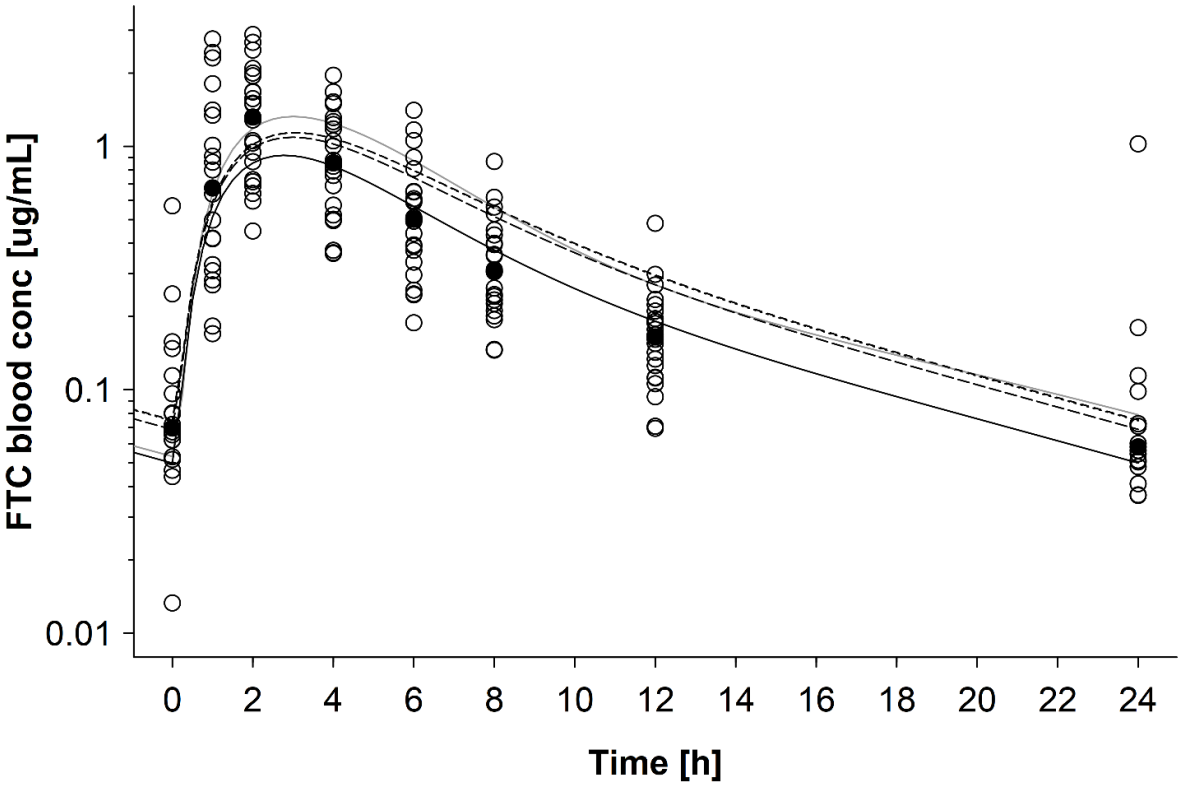


Figure S6: Concentration-time profile of acyclovir (**A**) and emtricitabine (**B**) in the umbilical vein in steady state after administration of oral acyclovir 400 mg state and oral emtricitabine 200 mg, respectively. Circles indicate observed *in vivo* data obtained at delivery and lines indicate predictions using different approaches for estimating placental drug permeability. For acyclovir, all observed concentration data were taken from the literature[15] and for emtricitabine, observed concentration data were both from the literature[12] and from the herein presented study. Dashed line: *ex vivo* cotyledon perfusion experiment; solid line: approach suggested by Zhang et al.[17]; dotted line: default method implemented in OSP. Further details can be found in the main manuscript.

A


ACV blood conc [ug/mL]
1
0.1
0.01
0
2
4
6
8
Time [h]

B:


FTC blood conc [ug/mL]
1
0.1
0.01
0
2
4
6
8
10
12
14
16
18
20
22
24
Time [h]

C:

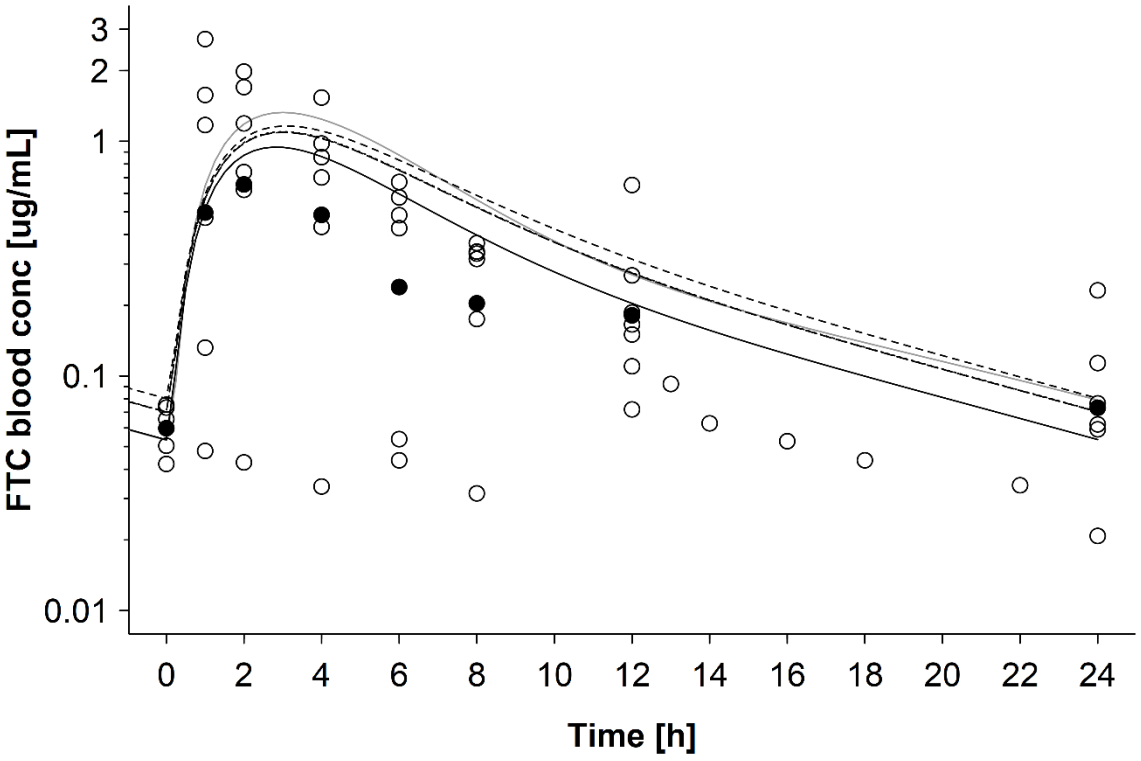


Figure S7: Sensitivity analysis of acyclovir (**A**) and emtricitabine (**B, C**) in steady state after administration of oral acyclovir 400 mg state and oral emtricitabine 200 mg, respectively. Empty circles indicate observed *in vivo* individual data obtained at steady state. Closed circles indicate the mean of the observed *in vivo* data. Lines indicate predictions using different approaches for estimating sensitivity of parameters. Grey solid line: non-pregnant model; black solid line: pregnant model; dotted line: pregnant model with non-pregnant parameters of GFRmat, fraction unbound, renal blood flow and kidney volume; dashed line: pregnant model with non-pregnant parameters of renal blood flow and kidney volume; long dashed line: pregnant model with non-pregnant parameters of kidney volume; For acyclovir, all observed concentration data were taken from the literature [ref] and for emtricitabine, observed concentration data were both from the herein presented study. Further details can be found in the main manuscript.

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